

MUHAMMAD TALHA

Associate Software Engineer | Data Scientist

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| [Portfolio](#)

SUMMARY

Computer Science graduate with practical experience in machine learning, backend development, and AI applications. Skilled in building end-to-end ML pipelines, deploying models with FastAPI and Docker. Interested in an entry-level Machine Learning, Data Science, or Software Engineering role.

TECHNICAL SKILLS

Programming: Python, SQL, C/C++, HTML, CSS
Machine Learning: Scikit-learn, PyTorch, Transformers, NLP
Backend: FastAPI, REST APIs, Pydantic
Data: Pandas, NumPy, Feature Engineering, EDA
Tools: Docker, Git, Linux, VS Code, Jupyter Notebook, Google Colab

EXPERIENCE

Machine Learning 1	Lahore, Pakistan
<i>Data Science Trainee</i>	Jun 2026 - Jul 2026
- Worked on real-world machine learning projects using Python and Scikit-learn. - Performed data preprocessing, exploratory data analysis (EDA), and feature engineering. - Built, trained, and evaluated classification models using cross-validation and hyperparameter tuning. - Collaborated using Git while following machine learning development workflows. - Gained hands-on experience in developing machine learning pipelines.	

PROJECTS

Next-Gen AI Desktop Assistant (Final Year Project)

- Developed an offline AI desktop assistant supporting voice and text interactions using local LLMs.
- Implemented modular task execution, file management, application and system control, and conversational workflows.
- Integrated long-term memory with ChromaDB to enable context-aware conversations while preserving user privacy.
- Tools:** Python, Ollama, Gemma 3, ChromaDB, PySide6, WhisperFlow, NLTK

Insurance Premium Category Predictor

- Built an end-to-end machine learning pipeline for insurance premium category prediction.
- Developed and deployed a REST API using FastAPI and Docker for scalable inference.
- Implemented structured logging, preprocessing pipelines, and error handling for production readiness.
- Tools:** FastAPI, Docker, Scikit-learn, Python, Linux

Bilingual Sentiment Analysis (Urdu & English)

- Fine-tuned a multilingual BERT (mBERT) model for Urdu-English sentiment classification.
- Built a custom preprocessing and tokenization pipeline to handle code-switched text.
- Trained and evaluated transformer models using the Hugging Face ecosystem and PyTorch.
- Tools:** Hugging Face Transformers, PyTorch, mBERT

EDUCATION

University of Engineering and Technology, Lahore , Lahore	2022 - 2026
BS in Computer Science	
Coursework: DB, ML, Data Science, NLP, AI	

CERTIFICATIONS

Machine Learning Specialization , Stanford / DeepLearning.AI (Coursera)	Jan 2025
Intermediate SQL , DataCamp	Jan 2025

RESEARCH EXPERIENCE

- Beyond the Cloud: Offline AI Desktop Assistant — 2026
- Co-authored a research paper on designing an offline-first AI desktop assistant using local large language models.
- Proposed a privacy-preserving architecture integrating local LLM inference, speech recognition, and modular command execution.
- Keywords:** Local LLMs, Edge AI, NLP, Speech Recognition, Privacy-Preserving AI